INCH-POUND

MIL-V-47311(MI)
29 December 1989

MILITARY SPECIFICATION

VALVE, PNEUMATIC EXHAUST

This specification is approved for use by the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines requirements for one type of pneumatic exhaust valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-35 Passivation Treatments for Corrosion Resisting Steel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

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MILITARY

MIL-M-9950	Missile Components, Liquid Oxygen, Liquid Nitrogen, Gaseous Oxygen, Gaseous Nitrogen, Instrument Air, Helium and Fuel Handling Systems, Cleaning and Packaging for Delivery

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-1190	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A276	Standard and Specification for Stainless and Heat-Resisting Steel Bars and Shapes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103. Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. Materials to be used in construction of the valve shall be selected in accordance with the order of precedence specified in MIL-STD-970.

3.2.1 Valve body and fitting. Valve body and actuation inlet fitting shall be corrosion resistant steel in accordance with ASTM A276, type 316 (see 4.6.1).

3.2.2 Finish. External surfaces of the valve shall be passivated in accordance with QQ-P-35 (see 4.6.1).

3.3 Design and construction. The design and construction of the valve shall conform to figures 1 and 2 of this specification (see 4.6.1).

3.3.1 Internal volume. Internal volume of the valve actuation system from inlet port to outlet port shall not exceed 0.04 cubic inches (cu. in.) after actuation (see 4.6.1).

3.3.2 Weight. Unit weight shall not exceed 0.05 pounds (lbs) (see 4.6.1).

3.4 Performance.

3.4.1 Actuation pressure and time. The exhaust poppet of the valve shall open with a pressure indication of 5 + 1 pounds per square inch gage (psig) at the exhaust outlet port within 100 milliseconds (ms) after application of an internal pressure of 70 ± 5 psig with a steady state application of 5 ± 1 psig at the exhaust inlet port (see 4.6.2).

3.4.2 Resealing pressure. The exhaust poppet of the valve shall reseal with no leakage when the actuation pressure is decreased from 60 ± 5 psig to 10 psig minimum (see 4.6.3).

3.4.3 Actuation pressure drop. When a steady state pressure of 500 ± 25 psig of gaseous nitrogen (GN_2) is applied to the actuation inlet port at a minimum flow rate of 0.14 pounds per hour (lb/hr), the static pressure drop from the actuation inlet port to the actuation outlet port shall not exceed 10 psig (see 4.6.4).

3.4.4 Exhaust pressure drop. When a steady state pressure of 5 ± 1 psig of GN_2 is applied at the exhaust inlet port with a flow rate of 0.14 ± 0.02 lb/hr, the pressure drop from the exhaust inlet port to the exhaust outlet port shall not exceed 0.5 psig (see 4.6.5).

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3.4.5 Flow rate. With GN_2 applied at the inlet port at an increasing rate of 10 ± 1 psig per minute, the flow rate at both the exhaust outlet port and the actuation outlet port of the valve shall be 0.14 ± 0.02 lb/hr before indication of a maximum actuation inlet pressure of 525 psig (see 4.6.6).

3.4.6 Leakage rate. With the actuation inlet port capped and the internal pressure of the valve established at 2.47 pounds per square inch absolute (psia) maximum, the pressure rise in the valve shall not exceed 0.15 pounds per square inch (psi) during a 15 minute period (see 4.6.7).

3.4.7 Safety relief pressure. With a pressure of 10 ± 5 psig applied to the exhaust inlet port and increased at a rate of 5 ± 1 psig per minute the valve poppet shall open and the pressure at the exhaust inlet port shall not exceed 25 psig (see 4.6.8).

3.4.8 Proof pressure. The valve shall withstand a pressure of 7100 ± 100 psig applied at the actuation inlet port with actuation outlet port sealed. The pressure shall be held for a minimum of 5 minutes. There shall be no leakage or permanent deformation (see 4.6.9).

3.4.9 Burst pressure. With the actuation outlet port sealed, the valve shall not rupture with application of 10,700 psig to the actuation inlet port (see 4.6.10).

3.5 Environmental requirements. The valve shall meet the requirements listed in table I while being subjected to the environmental conditions listed in subgroup I of table II and shall meet the requirements listed in table I after being subjected to the environmental conditions of subgroup II of table II.

TABLE I. Environmental performance requirements.

Performance characteristics	Requirement paragraph
Actuation pressure and time	3.4.1
Resealing pressure	3.4.2
Actuation pressure drop	3.4.3
Exhaust pressure drop	3.4.4
Flow rate	3.4.5
Leakage rate	3.4.6
Safety relief pressure	3.4.7

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TABLE II. Environmental conditions.

Environmental conditions	Condition paragraph	Method paragraph
Subgroup I		
High temperature	3.5.1	4.6.11
Low temperature	3.5.2	4.6.12
Subgroup II		
Thermal shock	3.5.3	4.6.13
Vibration	3.5.4	4.6.14
Shock	3.5.5	4.6.15

3.5.1 High temperature. The high temperature environment shall be an exposure to a temperature of 60 + 3° Celsius (C) [140 ± 5° Fahrenheit (F)] for 4.0 hours ± 12 minutes (see 4.6.11).

3.5.2 Low temperature. The low temperature environment shall be an exposure to a temperature of -43 ± 3°C (-45 ± 5°F) for 4.0 hours ± 12 minutes (see 4.6.12).

3.5.3 Thermal shock. The thermal shock environmental cycle shall be from room ambient conditions to 32 ± 3°C (90 ± 5°F) within one hour and hold for 4 hours + 12 minutes, then transfer to -54 ± 3°C (-65 ± 5°F) and hold for 4 hours ± 12 minutes and transfer to room ambient (see 4.6.13).

3.5.4 Vibration. The valve shall operate satisfactorily after being subjected to vibration along each of three mutually perpendicular axis at an intensity of 10 ± 1 gravity units (g) at a frequency of 50 ± 3 hertz for 60 ± 3 minutes (see 4.6.14).

3.5.5 Shock. The valve shall operate satisfactorily after being subjected to a shock pulse along each of three mutually perpendicular axis in both directions at an intensity of 400 + 0, -40 g at a maximum rise and decay time of 2.0 milliseconds (ms) (see 4.6.15).

3.6 Identification marking. Each valve shall be identified in accordance with figure 1 (see 4.6.1).

3.7 Workmanship. The valve shall be clean and free from loose splattered materials and other foreign matter (see 4.6.1).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.3.1, 4.3.2 and 4.3.3 of this specification.

4.3.1 Inspection equipment and facilities. The inspection equipment and facilities shall be of adequate quality and quantity to permit the performance of the required inspections.

4.3.2 Ambient temperature, pressure and humidity. Ambient temperature shall be 15°C (60°F) to 35°C (95°F); barometric pressure shall be local standard (corrected to 28 to 32 inches of mercury) and humidity shall be 10 to 95 percent.

4.3.3 Test medium. Unless otherwise specified, all inspections and tests shall be performed using a test medium of GN_2 at room ambient temperature with a dew point of not more than -57°C (-71°F).

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4.4 First article inspection. When specified by the procurement document the first article inspection shall consist of the examination and tests specified under 4.6 and shall be conducted in the order specified in 4.4.1. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.2).

4.4.1 First article sample. The first article sample shall consist of six valves, representative of the production product. The first article sample shall be subjected to all inspections and tests specified in 4.6 in the order shown.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of group A and B inspections.

4.5.1 Inspection lot. An inspection lot shall consist of all valves which are manufactured under uniform conditions and offered for inspection at any one time.

4.5.2 Group A inspection. Group A inspection shall consist of the inspections specified in table III, and shall be made on the same set of units in the order shown.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product--	3.2.1, 3.2.2, 3.3.1, 3.3.2, 3.6 & 3.7	4.6.1
Actuation pressure/time-	3.4.1	4.6.2
Resealing pressure-----	3.4.2	4.6.3
Actuation pressure drop-	3.4.3	4.6.4
Exhaust pressure drop---	3.4.4	4.6.5
Flow rate----------	3.4.5	4.6.6
Leakage rate----------	3.4.6	4.6.7
Safety relief pressure--	3.4.7	4.6.8
Proof pressure----------	3.4.8	4.6.9

4.5.2.1 Sampling plan. Quality conformance inspection shall be on a 100 percent basis. Sampling plans may be used for the product examination, exhaust pressure drop and proof pressure requirements, however, they must assure no less than 90 percent confidence that an entire lot of units is in compliance with these requirements must be submitted to the procuring activity for approval or must allow for the lots to be accepted when all of the items sampled from each perspective lot are in full compliance with the requirements of the parameter being inspected and rejected when one or more of the items sampled do not comply with the requirements.

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4.5.2.2 Rejection criteria. Failure of any unit or lot of units to comply with one or more of the characteristics specified in table III shall constitute a defective. In the event defectives are found the supplier shall take corrective action acceptable to the procuring activity to identify and correct the defectives and the cause of the defectives. When sample inspection is used and one or more defectives are found in the sample taken from a lot this lot shall be considered defective and shall be subject to 100 percent reinspection. When reinspection is performed only those items which are found defective shall be rejected from the lot. Acceptance of the product shall discontinue until this corrective action is implemented by the supplier and all units have successfully passed quality conformance inspection.

4.5.3 Group B inspection. Group B inspection shall consist of the inspections specified in table IV in the order listed. The procuring activity will consider alternate plans proposed by the supplier for demonstrating production quality under environmental conditions where the supplier has previously demonstrated adequate quality and is in continuous production. When no other plan is specified or authorized by the procuring activity, the inspections shall consist of those inspections specified in table IV.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
High temperature	3.5.1	4.6.11
Low temperature	3.5.2	4.6.12
Thermal shock	3.5.3	4.6.13
Vibration	3.5.4	4.6.14
Shock	3.5.51	4.6.15

4.5.3.1 Sampling plan. Sampling for group B inspection shall be on samples from lots which have passed group A inspection. Samples shall be taken by random selection from each month's production in accordance with the appropriate column in table V as specified by the procuring activity (see 6.2). After three consecutive months production have exhibited no failures when inspected as specified in table IV, the supplier may select the sample size in the next consecutively numbered column to the right of the column used in prior months. This procedure shall be repeated until column III in table VI is in effect. The acceptance number shall be zero.

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TABLE V. Group B sampling plan.

Monthly production rate	Monthly sampling size		
	Column I	Column II	Column III
0 to 250----------	5	4	3
251 to 750----------	4	3	2
751 to 1500----------	3	2	1

4.5.3.2 Noncompliance. If a sample fails to pass group B inspection, the lot shall be rejected and the supplier shall notify the procuring activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which are manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the procuring activity, has been taken. After the corrective action has been taken, group B inspections shall be repeated on additional sample units (all test and examinations, or the test which the original sample failed, at the option of the qualifying activity). Group A inspections may be reinstituted; however, final acceptance and shipment shall be withheld until the group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the procuring activity.

4.5.4 Inspection of packaging. Preservation, packing and marking shall be inspected for conformance to section 5.

4.6 Methods of inspection.

4.6.1 Examination of product. Each valve shall be visually and dimensionally inspected and weighed to determine conformance with the requirements of this specification (see 3.2.1, 3.2.2, 3.3.1, 3.3.2, 3.6 and 3.7).

4.6.2 Actuation pressure and time. A pressure of 5 ± 1 psig shall be applied to the exhaust inlet port. A pressure of 70 ± 5 psig shall be applied to the actuation inlet port. The time between application of pressure at the actuation inlet port and indication of 5 ± 1 psig at the exhaust outlet port shall be measured to determine conformance to 3.4.1.

4.6.3 Resealing pressure. A pressure of 5 ± 1 psig shall be applied at the exhaust inlet port. A pressure of 60 ± 5 psig shall be applied to the actuation inlet port. Opening of the valve shall be verified and the pressure at the actuation inlet port shall be gradually decreased until the valve reseals. The pressure at reseal shall be measured to determine conformance to 3.4.2.

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4.6.4 Actuation pressure drop. A steady-state pressure of 500 ± 25 psig shall be applied to the actuation inlet port of the valve at a minimum flow rate of 0.14 lb/hr. The pressure difference at the actuation outlet port shall be measured to determine conformance to 3.4.3.

4.6.5 Exhaust pressure drop. A steady-state pressure of 5 ± 1 psig shall be applied at the exhaust inlet port with a flow rate of 0.14 ± 0.02 lb/hr. The pressure difference at the exhaust outlet port shall be measured to determine conformance to 3.4.4.

4.6.6 Flow rate. A pressure of 15 psig shall be applied at actuation inlet port and gradually increased at a rate of 10 + 1 psig per minute while the exhaust inlet port pressure is maintained at 15 ± 5 psig until a flow rate of 0.14 ± 0.02 lb/hr is indicated at the exhaust outlet port. The actuation inlet pressure shall be measured at the initial indication of the specified flow rate to determine conformance to 3.4.5.

4.6.7 Leakage rate. The internal pressure of the valve shall be reduced and stabilized at 2.47 psia, maximum, the pressure rise shall be measured for a period of 20 ± 5 minutes to determine conformance to 3.4.6.

4.6.8 Safety relief pressure. A pressure of 10 ± 5 psig shall be applied to the exhaust inlet port, then gradually increased at a rate of 5 ± 1 psig per minute until the valve poppet opens and flow is indicated at exhaust outlet port. The exhaust inlet pressure at the initial indication of flow shall be measured to determine conformance to 3.4.7.

4.6.9 Proof pressure. The actuation outlet port shall be sealed and the actuation inlet port shall be pressurized to 7100 + 100 psig for 5 minutes. The valve shall be inspected for conformance to 3.4.8.

4.6.10 Burst pressure. The actuation outlet port shall be sealed and the actuation inlet port shall be pressurized to 10,700 psig. The valve shall be inspected for conformance to 3.4.9.

4.6.11 High temperature. The valve shall be subjected to the requirements of 3.5.1 and tested in accordance with 3.5, tables I and II.

4.6.12 Low temperature. The valve shall be subjected to the requirements of 3.5.2 and tested in accordance with 3.5, tables I and II.

4.6.13 Thermal shock. The valve shall be subjected to the requirements of 3.5.3 and tested in accordance with 3.5, tables I and II.

4.6.14 Vibration. The valve shall be subjected to the requirements of 3.5.4 and tested in accordance with 3.5, tables I and II.

4.6.15 Shock. The valve shall be subjected to the requirements of 3.5.5 and tested in accordance with 3.5, tables I and II.

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4.6.16 Inspection of packaging. Preservation, packing and marking shall be inspected for conformance to section 5.

5. PACKAGING

5.1 Preservation. Cleaning and preservation shall be in accordance with MIL-M-9950 to a liquid oxygen clean condition. All ports of the valve shall be capped to maintain the specified level of cleanliness.

5.2 Packing. Each valve shall be packed for shipment in accordance with MIL-STD-1190 applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve is intended for use as an exhaust valve in a cooling system.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Whether first article is required (see 3.1).

c. Alternate plan for group B inspection (see 4.5.3).

d. Applicable level of preservation, packing and marking if other than specified (see section 5).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of six units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.4 Subject term (key word) listing.

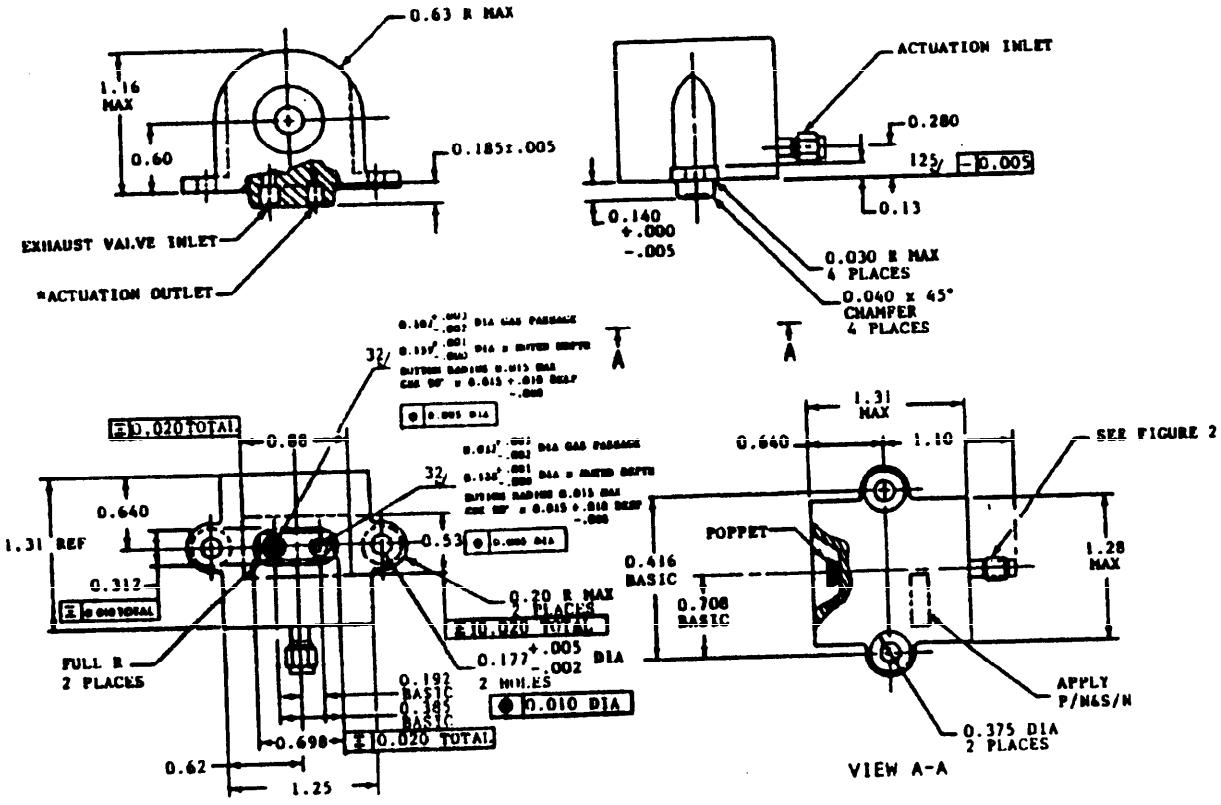
Valve, pneumatic
Exhaust valve, pneumatic

Custodian:
Army - MI

Preparing activity:
Army - MI

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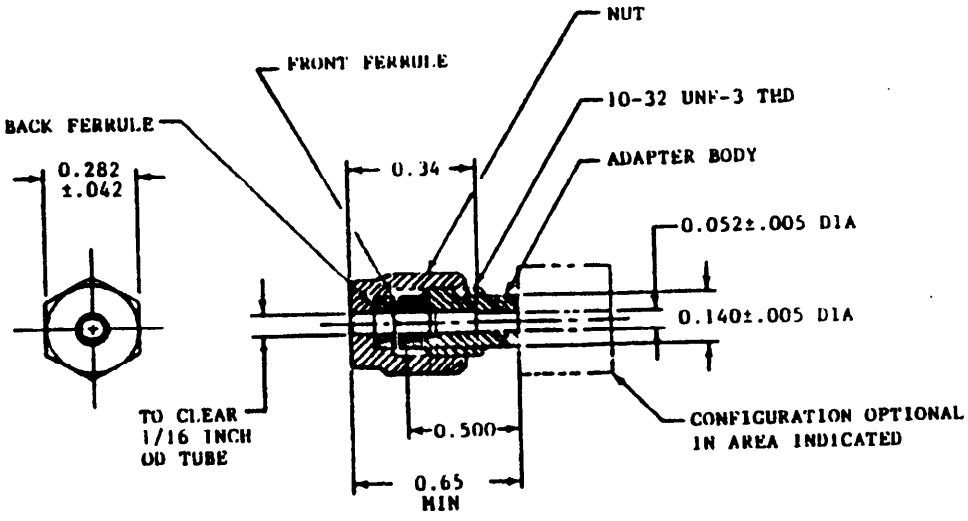


NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 1. Valve configuration.

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NOTES:
1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 2. Valve inlet fitting.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-V-47311(MI)	VALVE, PNEUMATIC EXHAUST

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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